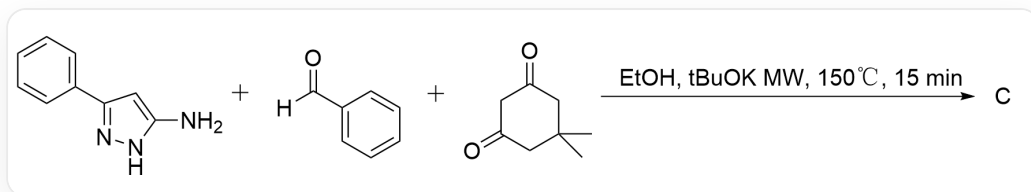


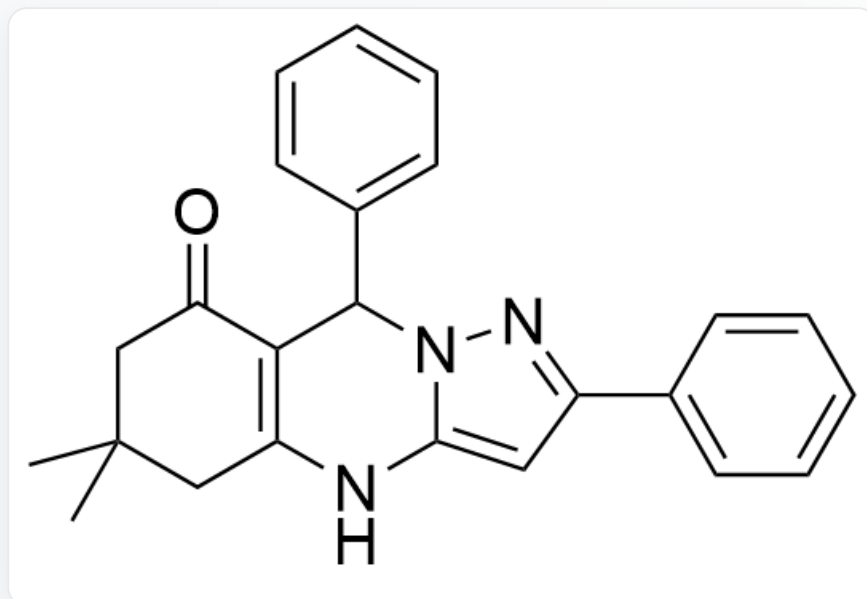
Question



NC1=CC(C2=CC=CC=C2)=NN1.O=C(C1=CC=CC=C1)[H].O=C1CC(C)(C)CC(C1)=O>EtOH, tBuOK
 MW, 150°C, 15 min>C

Without considering enantiomers, please predict the product C of this reaction (molecular formula: $C_{24}H_{25}N_3O_2$), and indicate how many rings are in this structure.

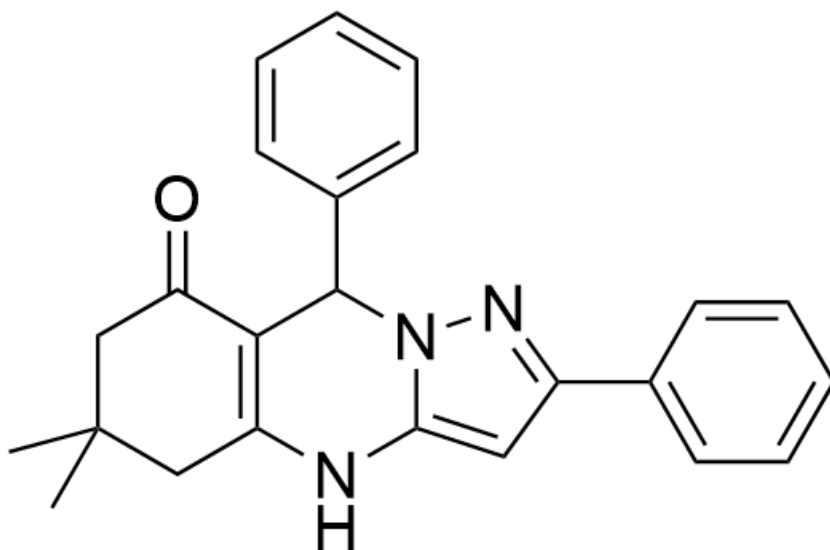
A.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

This structure contains 6 rings

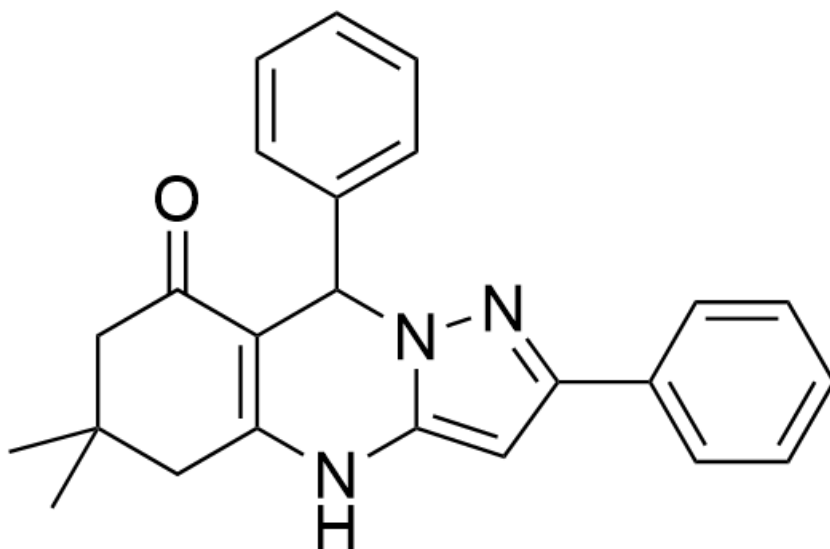
B.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

This structure contains 5 rings

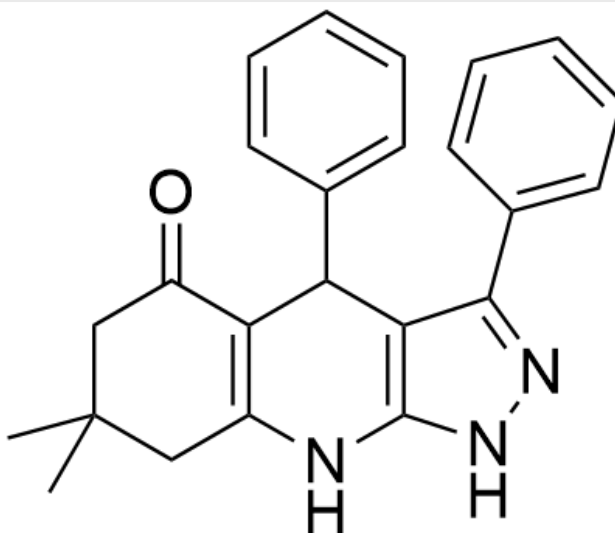
C.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

There are 8 rings in this structure

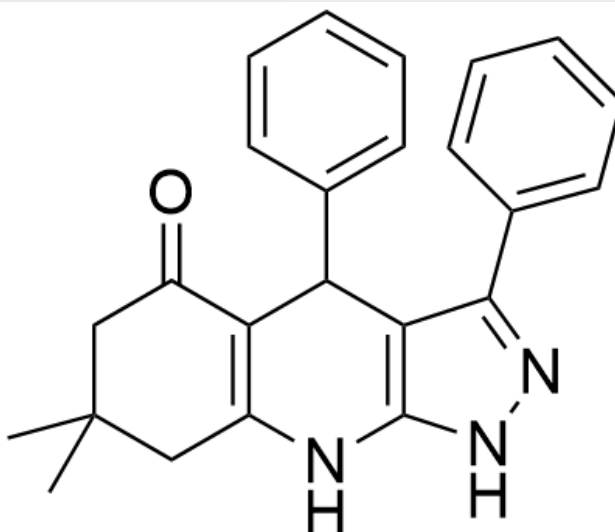
D.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

There are 6 rings in this structure

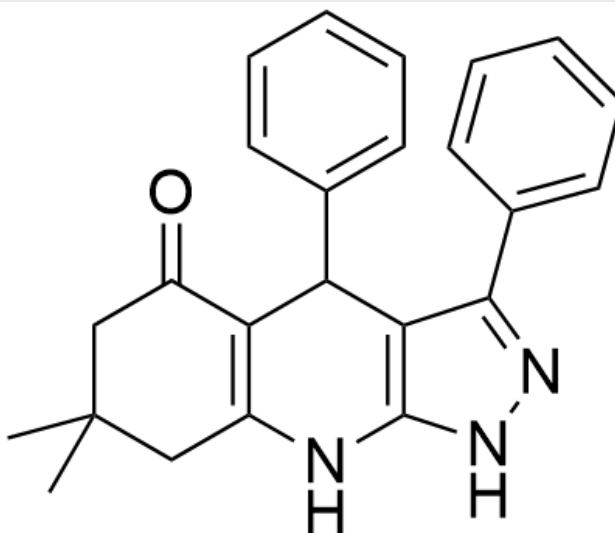
E.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

This structure contains 5 rings

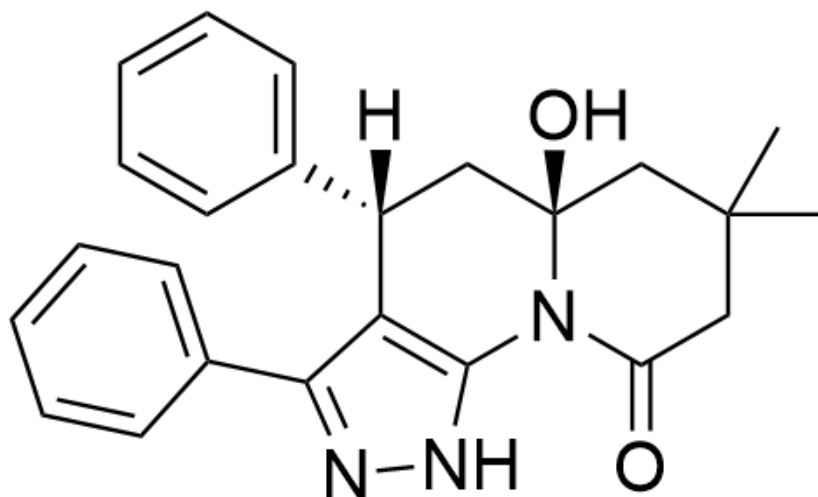
F.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

This structure contains 8 rings

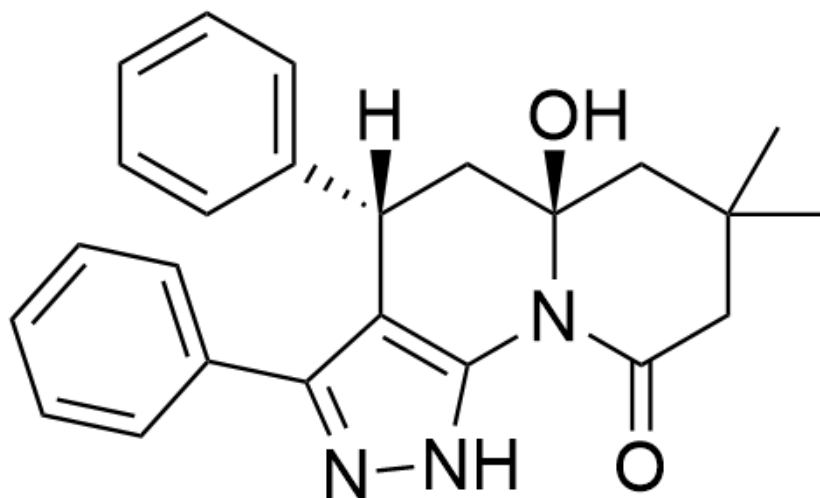
G.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 6 rings in this structure

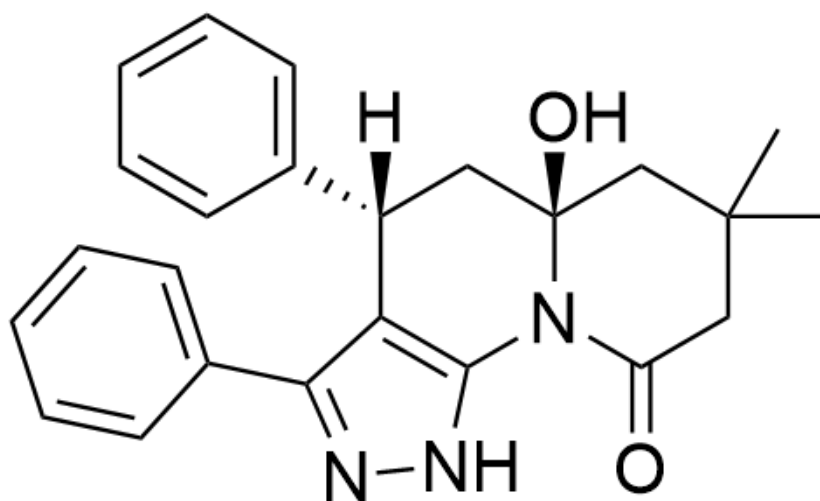
H.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 5 rings in this structure

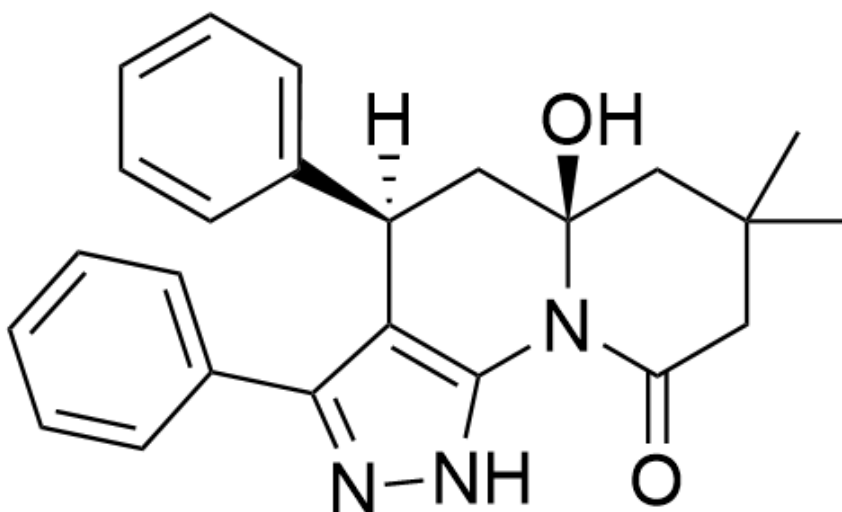
I.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 8 rings in this structure

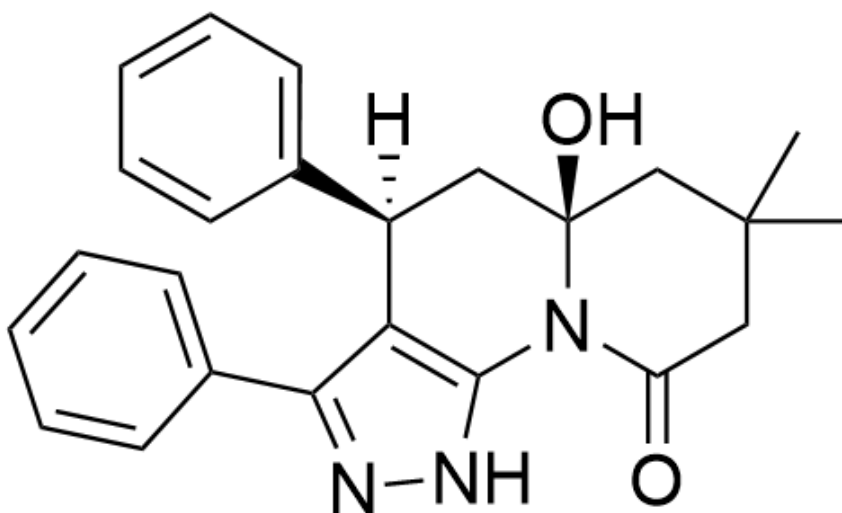
J.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

This structure contains 6 rings

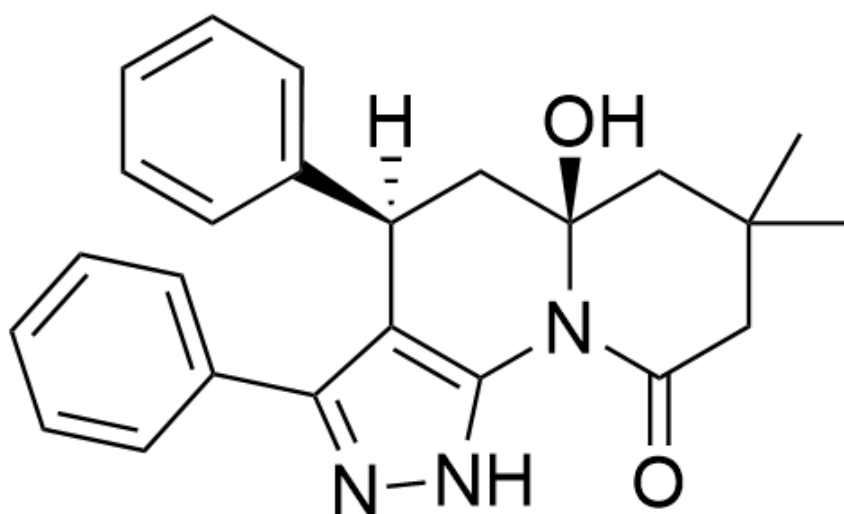
K.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 5 rings in this structure

L.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

This structure contains 8 rings

Answer

Correct Answer: **H**

Detailed Explanation

The reaction yields thermodynamically stable products under the condition of 150°C

CHECKPOINT

1 PTS

The reaction yields thermodynamically stable products under the condition of 150°C

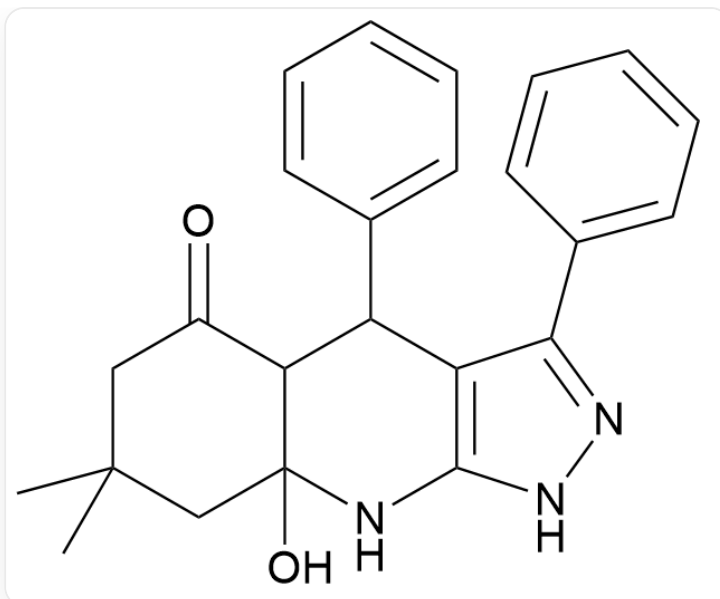
The carbon-carbon single bond is more stable than the carbon-nitrogen single bond during ring formation.

CHECKPOINT

1 PTS

The carbon-carbon single bond is more stable than the carbon-nitrogen single bond during ring formation

Thus, the reaction initially forms an intermediate.

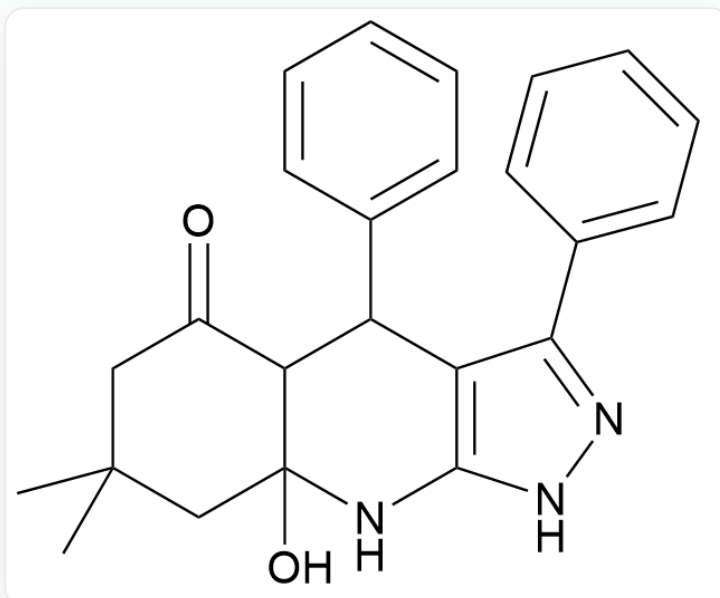


O=C1C2C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)(O)CC(C)(C)C1

CHECKPOINT

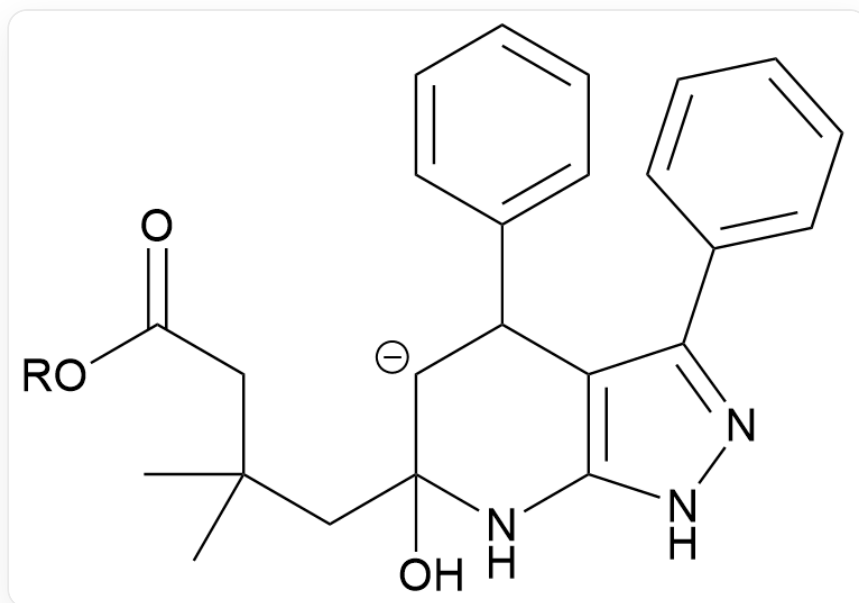
1 PTS

Thus, the reaction initially forms an intermediate



O=C1C2C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)(O)CC(C)(C)C1

Subsequently, under the action of a strong base, $tBuO^-$ or EtO^- nucleophilically attacks the carbonyl group, leading to ring opening.

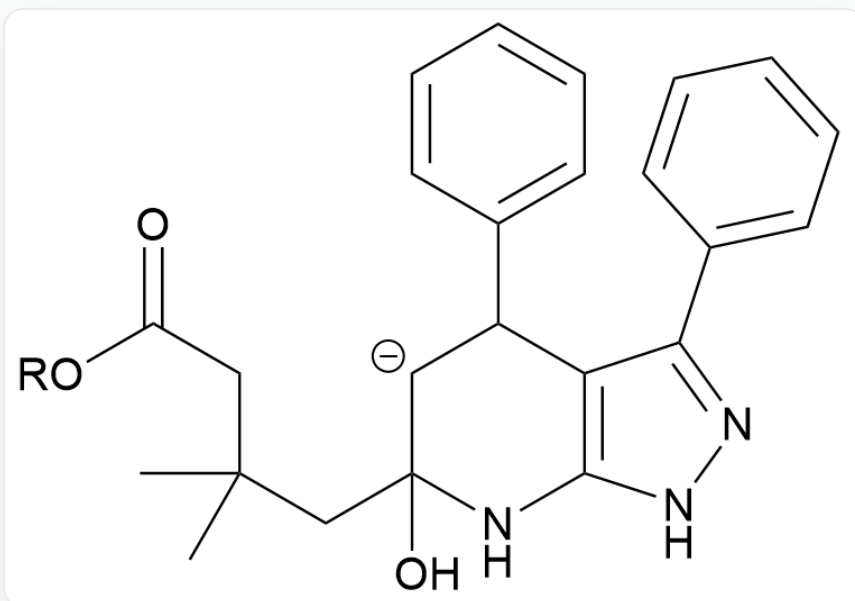


CC(CC1(O)[CH-]C(C2=CC=CC=C2)C3=C(NN=C3C4=CC=CC=C4)N1)(C)CC(O[R])=O

CHECKPOINT

1 PTS

Subsequently, under the action of a strong base, $tBuO^-$ or EtO^- nucleophilically attacks the carbonyl group, leading to ring opening


CC(CC1(O)[CH-]C(C2=CC=CC=C2)C3=C(NN=C3C4=CC=CC=C4)N1)(C)CC(O[R])=O

After proton exchange, further ring closure forms a stable lactam product.

CHECKPOINT

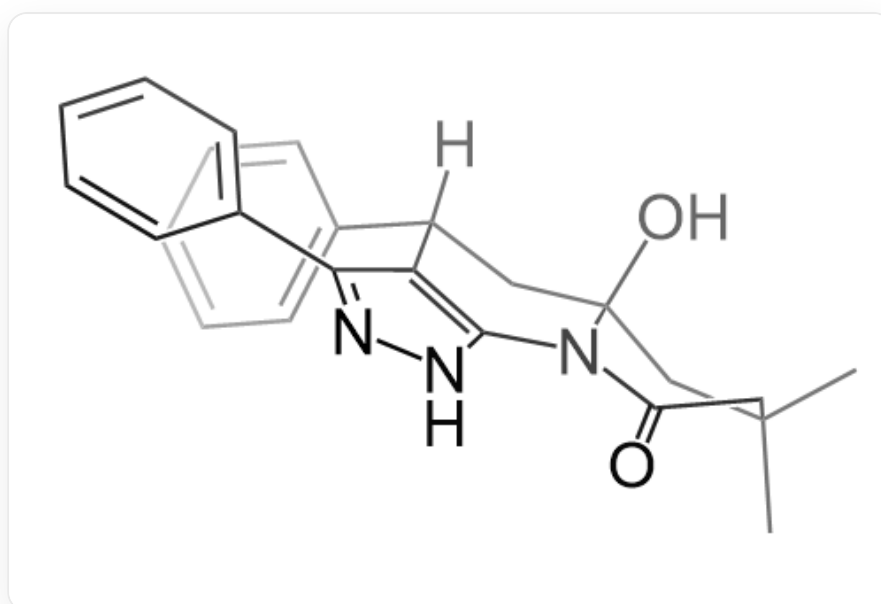
1 PTS

After proton exchange, further ring closure forms a stable lactam product

CHECKPOINT

1 PTS

The phenyl group in the equatorial position is the thermodynamically stable product



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1